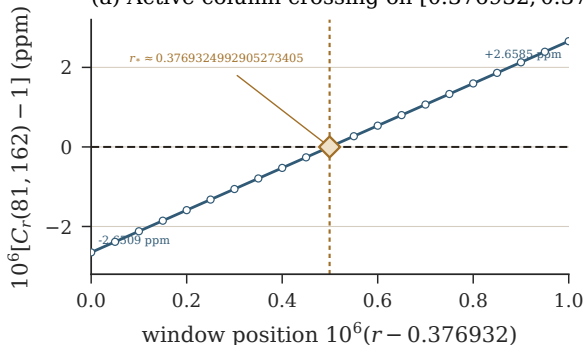




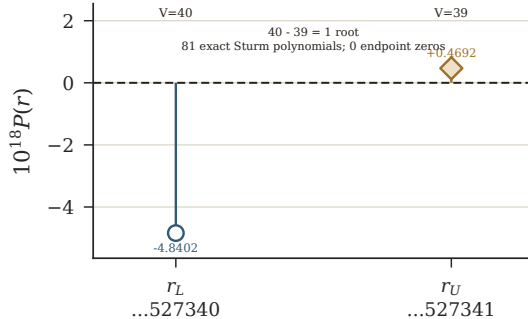
R0.48 exact threshold-root diagnostics

Exact GMP arithmetic; unique root of $P(r) = C_r(81, 162) - 1$ and full-window dominance in the current norm

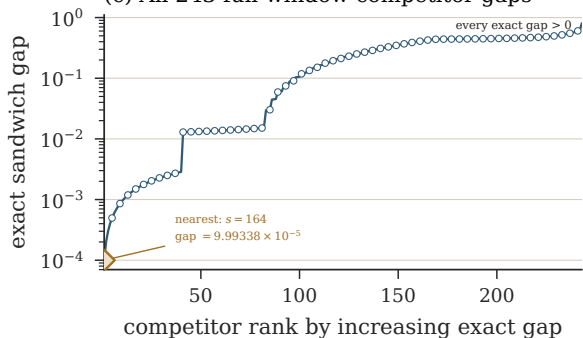
(a) Active-column crossing on $[0.376932, 0.376933]$



(b) Exact width- 10^{-18} root bracket



(c) All 243 full-window competitor gaps



Root: $0.376932499290527340 < r_* < 0.376932499290527341$; $P(r) > 0$ for every $r > 0$.

Dominance: 238 other fixed charges + 1 inactive endpoint + 4 remaining sectors = 243 exact competitors.

Threshold: the induced norm is < 1 , $= 1$, and > 1 below, at, and above r_* within the displayed window.

Scope: reduced canonical edge system and one weighted-l1 norm; this is not a PDE regularity or singularity theorem.

(d) Nearest monotone-sandwich bounds

